RESEARCHLY AI: AI-POWERED RESEARCH ASSISTANT AND COLLABORATION PLATFORM FOR UNIVERSITY STUDENTS

Group Thesis — Common Integrated Solution Report

Team member	Student ID	Individual component
Kariyawasam K.D.T.	IT22210524	Research Integrity and Compliance Module
Gunathilaka S.P.U.	IT22055194	Collaboration and Recommendation Engine
Hewamanne N.V.	IT22135902	Research Data Collection and Management Module
Jayasundara H.W.S.S.	IT22553096	Research Performance Analytics and Visualization Module

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DECLARATION AND SUPERVISOR CERTIFICATION

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Student name	Student ID	Signature	Date
Kariyawasam K.D.T.	IT22210524		
Gunathilaka S.P.U.	IT22055194		
Hewamanne N.V.	IT22135902		
Jayasundara H.W.S.S.	IT22553096		

Supervisor certification. The above candidates have carried out research for the bachelor's degree dissertation under my supervision.

Role and name	Signature	Date
Supervisor: Ms. Sanjeevi Chandrasiri		
Co-supervisor: Ms. Ishara Weerathunga		

ABSTRACT

Undergraduate research is supported by tools that were never designed to work together: students format references in one application, search literature in another, check similarity in a third and are allocated supervisors through manual processes, while institutions hold research output they cannot analyse. This report presents Researchly AI, an integrated AI-powered research assistant and collaboration platform built for SLIIT under project R26-IT-116. Four components operate behind one Next.js interface, an authenticated Express.js gateway and a shared Supabase PostgreSQL/pgvector data layer grounded in 4,219 records from SLIIT's institutional repository: research integrity and compliance (deterministic APA/IEEE citation formatting, evidence-backed gap analysis, retrieval-grounded proposal drafting, mind maps); collaboration and recommendation (fine-tuned Sentence-BERT supervisor matching with capacity-aware, explained ranking, peer groups, verified aspect-level feedback and effectiveness scoring); research data management (corpus quality audit, categorisation, faithful extractive summarisation, hybrid overlap analysis); and research analytics (ARIMA forecasting with prediction intervals, quality scoring and success prediction). A Design Science Research approach was followed and the platform is continuously deployed on Vercel and Railway with CPU-only inference.

The citation engine met its 90% field-accuracy target on 3,866 references, the supervisor matcher reached F1 0.970, and pairwise comparison and summarisation behaved correctly, whereas topic classification and corpus completeness fell short. Of 52 live validation cases signed off by the supervisor, 36 passed, 9 were partial, 6 failed and 1 was blocked. The work shows that grounding AI research support in an institution's own output yields explainable, citable assistance, and that shared data quality is the decisive factor for an integrated platform.

Keywords: research assistant, academic integrity, supervisor recommendation, research analytics, Sentence-BERT

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
APA	American Psychological Association
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
AUC	Area Under the (ROC) Curve
BERT	Bidirectional Encoder Representations from Transformers
CORS	Cross-Origin Resource Sharing
CPU	Central Processing Unit
DOI	Digital Object Identifier
DSR	Design Science Research
F1	Harmonic mean of precision and recall
GO	Group Objective
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
JWT	JSON Web Token
KPI	Key Performance Indicator
LLM	Large Language Model
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MMR	Maximal Marginal Relevance
NLP	Natural Language Processing
OTP	One-Time Password
PDF	Portable Document Format
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
RAG	Retrieval-Augmented Generation
RDA	Research and Digital Archive (SLIIT institutional repository)
REST	Representational State Transfer
RLS	Row-Level Security
SBERT	Sentence-BERT
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology

Abbreviation	Description
SP	Subproblem
SUS	System Usability Scale
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
UI	User Interface
XGBoost	Extreme Gradient Boosting

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1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Undergraduate research has become both more important and more difficult. Global scholarly output grew from roughly 2.1 million articles in 2010 to more than 3.3 million in 2022 [1], and a final-year student must survey that literature, choose a defensible topic, find a suitable supervisor, cite sources correctly, avoid overlap with earlier work and write a structured proposal, typically within a single academic year. In Sri Lanka, as elsewhere, research-based curricula are expected to develop exactly these skills [2], yet the tools that support them were never designed to work together: references are formatted in one application, literature is searched in a second, similarity is checked in a third and supervisors are allocated through departmental spreadsheets [3], [4].

Each fragment has a mature literature. Reference parsers such as CERMINE extract bibliographic metadata but do not render submission-ready citations [5]; large-language-model agents generate research ideas but do not provide evidence that a gap exists [6], [7]; Sentence-BERT made semantic similarity between texts cheap enough to compute at scale [8]; academic recommenders recommend papers far more often than people [4]; plagiarism detectors handle verbatim copying well but paraphrase poorly and operate after submission [3]; institutional repositories store research for deposit rather than analysis [9]; and bibliometric tools describe past literature but do not forecast an institution's own trajectory [10], [11]. What is missing is an integrated system in which the output of one stage is available to the next and every recommendation is grounded in the institution's own research.

This report presents **Researchly AI**, the outcome of research project R26-IT-116: an AI-powered research assistant and collaboration platform for university students, built for the Sri Lanka Institute of Information Technology (SLIIT). The platform integrates four components — research integrity and compliance, collaboration and recommendation, research data management, and research performance analytics — behind a single web application and a shared data substrate grounded in 4,219 records from SLIIT's institutional repository.

1.2 Stakeholders and User Needs

Table 1.1: Stakeholders and confirmed needs

Stakeholder	Role	Primary needs
Undergraduate and postgraduate researchers	Primary users	Correct citations; evidence-backed topic and gap; grounded proposal draft; early overlap warning; suitable supervisor and collaborators; quick summaries
Supervisors	Primary users	Well-matched students; proposals grounded in prior work; early warning of weak projects; fair feedback
Research coordinators and heads of department	Secondary users	Transparent allocation; supervision-quality view; topic trends
Library / repository staff	Data stewards	Clean, de-duplicated, analysable corpus
SLIIT (institution)	Owner	Academic integrity; quality and visibility of research output; data protection compliance

These needs were derived from the four individual proposals, supervisor discussions during the progress presentations and the validation reviews in August 2026. Two personas guided design: a final-year student starting from a vague idea with limited supervision time, and a coordinator who must allocate supervisors and monitor progress across many projects.

1.3 Alignment with the Sustainable Development Goals

The platform contributes primarily to **SDG 4 (Quality Education)** — target 4.3 on equal access to quality tertiary education and target 4.4 on relevant skills — by giving every student structured, explainable research support regardless of personal networks or paid tools. It contributes to **SDG 9 (Industry, Innovation and Infrastructure)**, target 9.5, by turning an institutional repository into queryable research infrastructure [12]. Collaboration features also support partnership building within the institution (SDG 17). The contribution is plausible rather than demonstrated: no longitudinal outcome study has yet been performed (Section 7.5).

1.4 Purpose and Scope of the Common Report

This common report presents the integrated problem, the complete solution, its integration, shared infrastructure, team execution and system-level evaluation. Detailed research methods, component designs and experiments are reported in the

four individual theses, referenced here as the Module 1–4 theses. Table 1.2 lists the team and components.

Table 1.2: Team members and individual components

Team member	Student ID	Individual component
Kariyawasam K.D.T.	IT22210524	Module 1: Research Integrity and Compliance Module
Gunathilaka S.P.U.	IT22055194	Module 2: Collaboration and Recommendation Engine
Hewamanne N.V.	IT22135902	Module 3: Research Data Collection and Management Module
Jayasundara H.W.S.S.	IT22553096	Module 4: Research Performance Analytics and Visualization Module

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

University students and institutions lack an integrated, explainable research-support system that grounds recommendations in the institution's own research output. Students move between disconnected tools for referencing, topic justification, supervisor selection, overlap checking and quality assessment, and the institution cannot see its research output as an analysable corpus. The result is lost time, avoidable citation and integrity errors, poorly matched supervision and decisions made without evidence.

2.2 Root Causes and Overall Research Gap

- **Fragmented tooling:** each task is served by a separate product; information produced at one stage (concepts, topics, drafts) is unavailable to the next [4], [3].
- **Generic evidence:** existing AI tools draw on global web content or free generation, so their outputs cannot be traced to citable local work [13].
- **Missing people-matching:** academic recommenders focus on papers; supervisor allocation relies on keyword lists [14].
- **Unanalysable repositories:** institutional data is incomplete, duplicated and uncategorised [9], [15].
- **No institutional foresight:** analytics are descriptive and citation-based, which is meaningless for student work [16].

The overarching gap (G1 in the group research paper) is the absence of a platform in which citation support, collaboration matching, document analysis and research analytics operate over a shared, institution-specific data substrate. The remaining gaps follow from it, because a fragmented toolchain gives no component access to what another has produced.

2.3 Subproblem and Component Mapping

Table 2.1: Subproblems and the components that address them

Subproblem	Component and owner	Input → output	Why necessary
SP1 Students cannot easily produce correct citations, evidenced gaps, grounded proposals and concept maps.	Module 1 — Research Integrity and Compliance (Kariyawasam K.D.T.)	References, topic, text → APA/IEEE citations, ranked gaps, proposal, mind map	Supports the writing stage; uses the corpus as evidence
SP2 Students cannot identify suitable supervisors and collaborators; supervision feedback is unanalysed.	Module 2 — Collaboration and Recommendation (Gunathilaka S.P.U.)	Proposal, ratings → ranked supervisors, groups, aspect sentiment, effectiveness	Supports the people side of research
SP3 Institutional research output is not available as a clean, categorised, summarised, similarity-aware corpus.	Module 3 — Research Data Collection and Management (Hewamanne N.V.)	SLIIT corpus, papers → categories, summaries, overlap analysis, quality metrics	Provides the shared data substrate for all modules
SP4 Students and institutions lack forecasts and timely, explainable feedback on research quality and risk.	Module 4 — Research Performance Analytics (Jayasundara H.W.S.S.)	Corpus, proposals → forecasts, quality and success scores	Supports topic choice and early intervention

The four subproblems are non-overlapping but interdependent: Module 3's corpus is the evidence base for Modules 1 and 4 and the training source for their models; Module 1's refined topic and proposal are the natural inputs to Module 2's matching and Module 4's scoring.

2.4 Overall Aim and Group Objectives

Aim. To design, implement and evaluate an integrated, AI-powered research assistant and collaboration platform that supports university students from topic selection to proposal submission, grounded in the institution's own research output.

Table 2.2: Group objectives, components and evidence

Group objective	Components	Evaluation evidence
GO1 Provide a single authenticated web platform giving access to all research-support features	All	Deployed system; end-to-end flow (Section 6.2)
GO2 Build and maintain a shared, embedded institutional corpus used by all components	M3 (provider), M1, M4	Corpus audit; index reuse (Section 6.3)
GO3 Ground recommendations in identifiable SLIIT research	M1, M2, M3	Traceable gaps, proposals, related papers, matches

Group objective	Components	Evaluation evidence
GO4 Meet component-level accuracy targets for the core models	M1–M4	Benchmarks in Table 6.1
GO5 Validate every component through live functional tests with supervisor sign-off	M1–M4	52 test cases (Figure 6.1)
GO6 Deploy continuously on cloud infrastructure with CPU-only inference	All	Vercel, Railway, GitHub Actions (Section 5.3)

2.5 Scope, Assumptions and Constraints

- **Scope:** English-language research support for SLIIT students and staff; the SLIIT RDA corpus plus arXiv and Semantic Scholar; web access on desktop and mobile.
- **Exclusions:** automatic allocation or grading decisions; non-English text; native mobile apps.
- **Assumptions:** repository metadata is broadly accurate; users review all generated content; supervisor directory data is current.
- **Constraints:** CPU-only hosting within a student budget (approximately LKR 115,000–131,000 per component in the proposals); limited labelled data; external API rate limits; a fixed academic timeline (Figure 5.1).

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

Researchly AI presents one web application through which a student moves from idea to proposal. The central value proposition is that **every output is grounded and explainable**: citations follow deterministic style rules, gaps and proposal fragments cite real SLIIT papers, supervisor matches show similarity and reasons, summaries are extractive, and analytics expose their components. A shared design principle runs through all four modules — deterministic methods where rules exist, retrieval where evidence exists, and generation only where interpretation is genuinely required — combined with a three-tier fallback (local model → Gemini → deterministic heuristic) so that the product never hard-fails. Figure 3.1 shows the end-to-end journey.

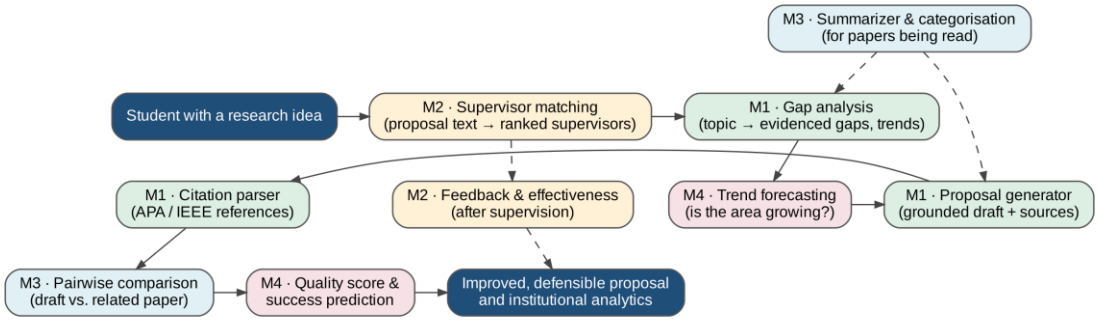


Figure 3.1: End-to-end student journey across the four modules

3.2 High Level Architecture

Figure 3.2 shows the four-layer architecture. The presentation layer is a Next.js 14 application on Vercel. The integration layer is an Express.js API gateway on Railway that verifies Supabase-issued JSON Web Tokens [17], applies rate limiting and proxies requests to the intelligence layer with module-specific timeouts — the API-gateway pattern for micro-services [18], [19]. The intelligence layer consists of four FastAPI services (ports 8001–8004) and a paper-chat service. The data layer is Supabase: PostgreSQL with pgvector [20] (30 tables with Row-Level Security [21]), authentication and storage, plus a shared SLIIT corpus index of 4,219 papers with 384-dimensional SBERT embeddings. External services are Gemini 2.5 Flash [22], CrossRef [23], Semantic Scholar [24], SLIIT RDA and arXiv.

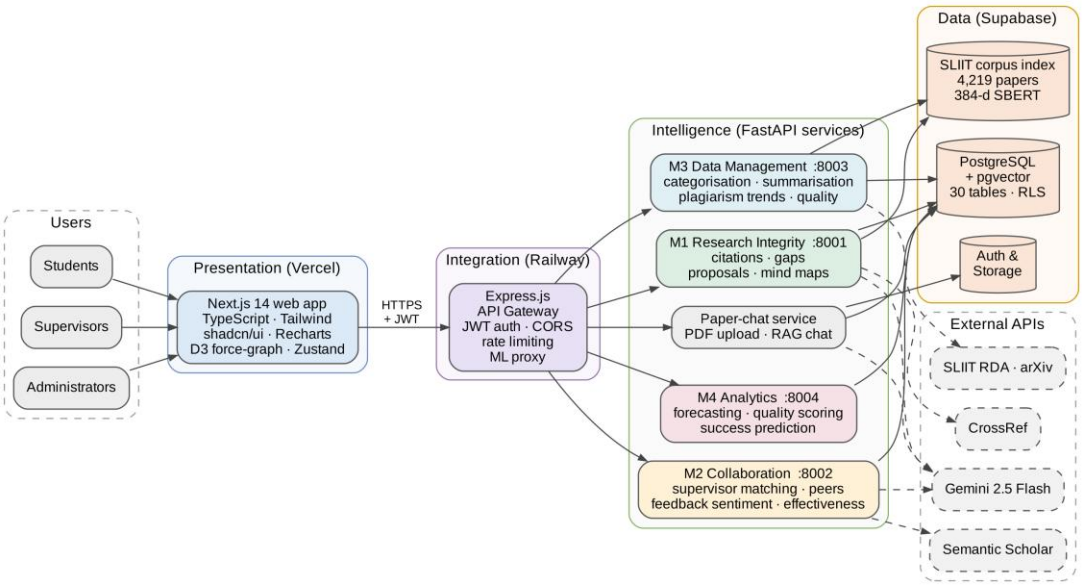


Figure 3.2: High-level architecture of the integrated Researchly AI platform

3.3 Component 1 Summary: Research Integrity and Compliance

Owner: Kariyawasam K.D.T. (IT22210524). **Subproblem:** SP1. **Input:** raw references, DOIs, topics, pasted papers or PDFs, draft text. **Core methods:** a deterministic regex citation engine with APA/IEEE formatting and CrossRef lookup; SBERT retrieval over 553 author-stated gap sentences with multi-dimensional ranking and an intelligence layer (gap type, trend, cross-domain opportunities, saturation, directions); retrieval-grounded proposal composition over 3,858 SLIIT exemplars with a Gemini fallback; Gemini-assisted mind maps. **Output:** formatted citations, evidenced gaps, grounded proposal drafts, concept graphs. **Contribution:** supports the writing stage with outputs a student can cite and defend.

3.4 Component 2 Summary: Collaboration and Recommendation

Owner: Gunathilaka S.P.U. (IT22055194). **Subproblem:** SP2. **Input:** proposal text, ratings and written feedback. **Core methods:** a Sentence-BERT bi-encoder fine-tuned with contrastive loss on 147 labelled pairs, ranking 74 SLIIT supervisors by 0.70 × similarity + 0.30 × availability with template explanations; Semantic Scholar profile enrichment with a Gemini focus summary; a research-group marketplace; OTP-verified ratings with four-aspect Gemini sentiment; a transparent re-normalised effectiveness score. **Output:** ranked, explained supervisor matches, groups, aspect

sentiment and effectiveness. **Contribution:** connects students to the right people and closes the supervision feedback loop.

3.5 Component 3 Summary: Research Data Collection and Management

Owner: Hewamanne N.V. (IT22135902). **Subproblem:** SP3. **Input:** the SLIIT repository corpus; pasted papers and PDFs. **Core methods:** a pre-computed 4,219 × 384 SBERT index; a corpus quality audit; TF-IDF one-versus-rest multi-label categorisation; extractive summarisation with centroid scoring and MMR [25]; plagiarism-trend buckets and pairwise comparison combining SBERT (75%) and 4-gram Jaccard (25%). **Output:** the shared corpus index, categories, faithful summaries, overlap analysis and quality metrics. **Contribution:** provides the data substrate on which the other modules depend.

3.6 Component 4 Summary: Research Performance Analytics and Visualization

Owner: Jayasundara H.W.S.S. (IT22553096). **Subproblem:** SP4. **Input:** the corpus, titles and abstracts or uploaded files. **Core methods:** per-topic ARIMA(1,1,1) forecasts with 95% prediction intervals [26], emergence and recommendation scores; a four-dimension quality score blending XGBoost regressors [27] with calibrated heuristics; success prediction blending an XGBoost classifier (40%) with a heuristic (60%). **Output:** forecasts, scores, risk bands. **Contribution:** informs topic choice and enables early intervention.

3.7 End to End User and Data Flow

A typical journey (Figure 3.1) begins when a student describes an idea. Module 2 ranks supervisors whose expertise matches the description. Module 1 analyses the topic against SLIIT gap statements and shows trends; Module 4 indicates whether the area is growing. Module 1 drafts a proposal grounded in the most similar SLIIT papers and formats its references; Module 3 compares the draft with closely related papers; and Module 4 scores its quality and success probability. While reading, the student uses Module 3's summariser and categoriser. After supervision, Module 2 collects verified feedback. Data flows in the opposite direction too: Module 3's corpus and embeddings are consumed by Modules 1 and 4.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

Integration followed an interface-first, gateway-centred strategy. The monorepo (pnpm workspaces and Turborepo) was scaffolded with the front end, gateway, four service skeletons and the Supabase schema before component work began (April 2026), so that every module was reachable through the same authenticated path from the first sprint. Each module owner implemented their service behind a stable route prefix; the integration lead (Module 1 owner) maintained the gateway, shared libraries (`services/shared`: configuration, Supabase client, embedding utilities, Gemini client, preprocessing), CI/CD and deployment. Risks were managed through three mechanisms: shared models and indices were pre-computed and baked into images to avoid start-up failures; every service implemented fallback tiers; and each module exposed status endpoints that the dashboard's model-status grid displays.

4.2 Interfaces, Data Contracts and Shared Models

All client requests use `/api/v1/<feature>` on the gateway; the gateway forwards them to the owning service with `callMlService(moduleId, path, method, body)`. Table 4.1 summarises the interfaces; the full endpoint list is in Appendix A.

Table 4.1: Integration interfaces and data contracts

Interface	Producer → consumer	Contract
REST via gateway	Front end → Modules 1–4	JSON over HTTPS; Bearer JWT; 60 s timeout (150 s for Module 3); errors normalised to {error|message|detail}
Shared corpus	Module 3 → Modules 1, 4	papers_raw_sliit.json records {id, title, authors[], abstract, year, url, subject[]}; paper_embeddings.npy (4,219 × 384, float32, L2-normalised)
Shared encoder	Modules 1/3 (sbert_plagiarism); Module 2 (fine-tuned matcher)	all-MiniLM-L6-v2 architecture, 384-d; base checkpoint as fallback
research_papers table	Module 3 (writer)	keywords[] written by categoriser
Identity	Supabase Auth → all	profiles.id = auth.users.id; RLS policies per table

Figure 4.1 shows the core entities of the shared schema (13 migrations, 30 tables) grouped by owning module.

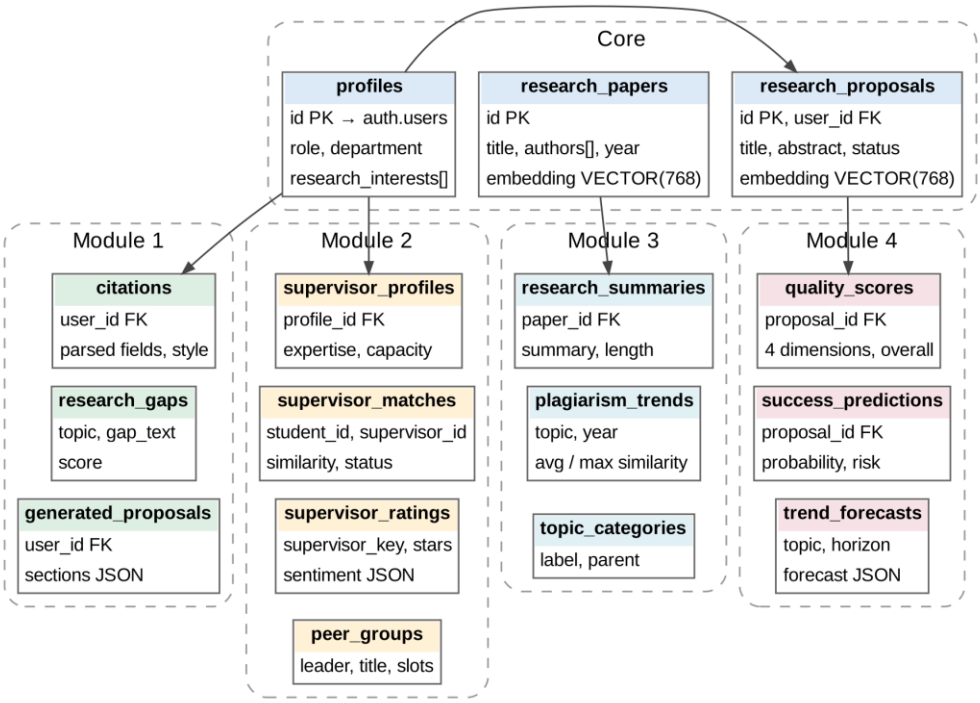


Figure 4.1: Simplified entity-relationship view of the shared Supabase schema

A representative data contract is the supervisor-matching response consumed by the front end:

```
POST /api/v1/matching/supervisors   { "proposal": "...", "top_k": 5 }
→ { "matches": [ { "supervisor_id": 71, "name": "...", "department":
"CSNE",
      "research_interests": ["Computer Networks", "Cyber Security"],
      "similarity_score": 0.8597, "multi_factor_score": 0.9018,
      "explanation": "...", "availability": true,
      "current_students": 1, "max_students": 3 } ] }
```

4.3 Shared Nonfunctional Requirements

Table 4.2: Shared non-functional requirements and how they are met

Quality attribute [28]	Requirement	Platform mechanism
Security	Authenticated access; least privilege; abuse protection	Supabase JWT verified at gateway; RLS on all tables; 600 req/min global and 200 req/min ML rate limits; CORS allow-list; security headers on Vercel [29]
Privacy	PDPA-aligned handling of personal data	Purpose-limited profile data; private storage bucket; OTP for ratings [30]
Performance	Interactive response on CPU	Pre-computed embeddings and model bundles; single matrix products for retrieval; sub-millisecond citation and scoring paths
Reliability	No hard failure in the UI	Three-tier fallbacks; status endpoints

Quality attribute [28]	Requirement	Platform mechanism
Maintainability	Independent evolution of modules	Micro-services with stable prefixes; shared library; typed interfaces
Usability and accessibility	Consistent, responsive interface	Common dashboard shell, light/dark themes, mobile slide-in sidebar, sample inputs, downloadable reports [31]
Explainability	Users can see why	Similarities, weights, sources and model versions returned in responses

4.4 Integrated Workflow and Exception Handling

Figure 4.2 shows the normal path of a request and the fallback decision. Exception paths are handled at three levels. At the **gateway**, missing or invalid tokens return 401 and rate-limit breaches return 429 with a message; upstream timeouts and errors are mapped to clear JSON errors. In each **service**, validation errors return 400 (for example, an abstract shorter than 50 characters), missing resources 404, and model or external-API failures trigger the next fallback tier (local model → Gemini → deterministic heuristic). In the **front end**, errors are shown inline next to the form that caused them. Validation identified places where degradation was too silent — Gemini parse failures in mind maps and aspect sentiment — and these are listed as integration risks (Section 6.6).

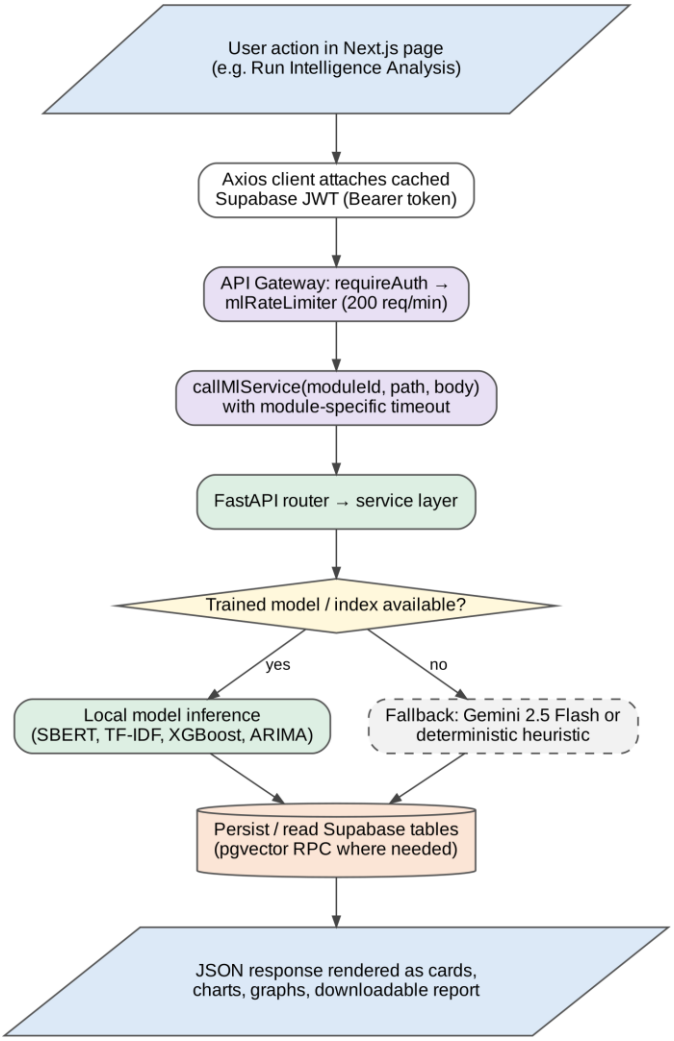


Figure 4.2: Request flow through the gateway, service and fallback tiers

4.5 Team Contribution and Dependency Matrix

Table 4.3 summarises each student's contribution, which is detailed in the individual theses and confirmed in Appendix B.

Table 4.3: Summary of each student's contribution

Member	Research and development	Integration and shared work	Depends on / provides
Kariyawasam K.D.T. IT22210524	Module 1: citation engine, gap analysis and intelligence layer, proposal retriever, mind maps; citation benchmark	Monorepo, API gateway, CI/CD, Vercel/Railway/Azure deployment; shared library; validation harness	Uses M3 corpus and encoder; provides topics/proposals to M2, M4

Member	Research and development	Integration and shared work	Depends on / provides
Gunathilaka S.P.U. IT22055194	Module 2: supervisor dataset and pairs, SBERT fine-tuning, multi-factor matching, profile enrichment, peer groups, OTP ratings, aspect sentiment, effectiveness	Migration 013 (peer groups, ratings); collaboration UI	Uses M1 topics; records matches
Hewamanne N.V. IT22135902	Module 3: corpus index, quality audit, topic classifier, extractive summariser, plagiarism trends and comparison	Shared corpus and embeddings; derived datasets for M1; data-management UI	Provides corpus to M1, M4
Jayasundara H.W.S.S. IT22553096	Module 4: ARIMA forecasting, quality and success models and heuristics, discipline classifier	Analytics UI	Uses M3 corpus; scores M1 proposals

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

Table 5.1: Common technology stack

Layer	Technologies	Justification
Presentation	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, shadcn/ui, Recharts, D3 / react-force-graph-2d, Zustand, Axios	Server-side rendering, typed components, responsive layout and rich interactive charts in one framework [32]
Integration	Express.js API gateway (TypeScript), express-rate-limit, Supabase JWT verification, per-module proxy with 60 s / 150 s timeouts	Single entry point that authenticates once and isolates the front end from service locations [18]
Intelligence	Python 3.10+, FastAPI, sentence-transformers (all-MiniLM-L6-v2, fine-tuned), scikit-learn, XGBoost, statsmodels, PyMuPDF / pypdf / pdfplumber, Google Gemini 2.5 Flash client	Asynchronous typed micro-services; CPU-friendly models that fit the hosting budget [33], [34]
Data	Supabase PostgreSQL 15 + pgvector, Row-Level Security, Supabase Auth and Storage; pre-computed NumPy embedding matrices and pickled model bundles	Relational and vector search in one managed service, policy-based access control [20], [21]
DevOps	pnpm workspaces + Turborepo monorepo, Docker multi-stage images, GitHub Actions, Vercel, Railway, Azure Container Registry workflow, Playwright	Reproducible builds and automatic deployment on every push to main [35]

Major choices were driven by two constraints: a student hosting budget, which ruled out GPU inference in production, and the need for explainable outputs, which favoured retrieval, deterministic rules and transparent blends over opaque generative models. Supabase was chosen because it provides relational data, vector search, authentication, storage and Row-Level Security in one managed service.

5.2 Development and Integration Process

The team worked in two-week sprints following Scrum practice [36] within the Design Science Research cycle [37], with tasks tracked in MS Planner, meetings and decisions recorded in MS Teams, and code in a single GitHub repository. Figure 5.1 shows the project timeline reconstructed from the proposal schedule and the repository history. Development phases visible in the commit log were: analytics service scaffolding (10 April 2026); corpus data collection (11 April); model wrappers (11 April); module

pages for citations, collaboration and data management (12 April); testing scaffolding with Playwright, Docker and seeding (14 April); paper upload and RAG chat (16 April); security hardening with RLS and removal of secrets (18–27 April); trained models for Modules 1, 2 and 4 (27 April–5 May); gap intelligence, peer connect and analytics endpoints (5–7 May); production deployment and performance fixes (13–14 May); profile enrichment and Azure container workflow (11–14 August); and validation reports (27–30 August).

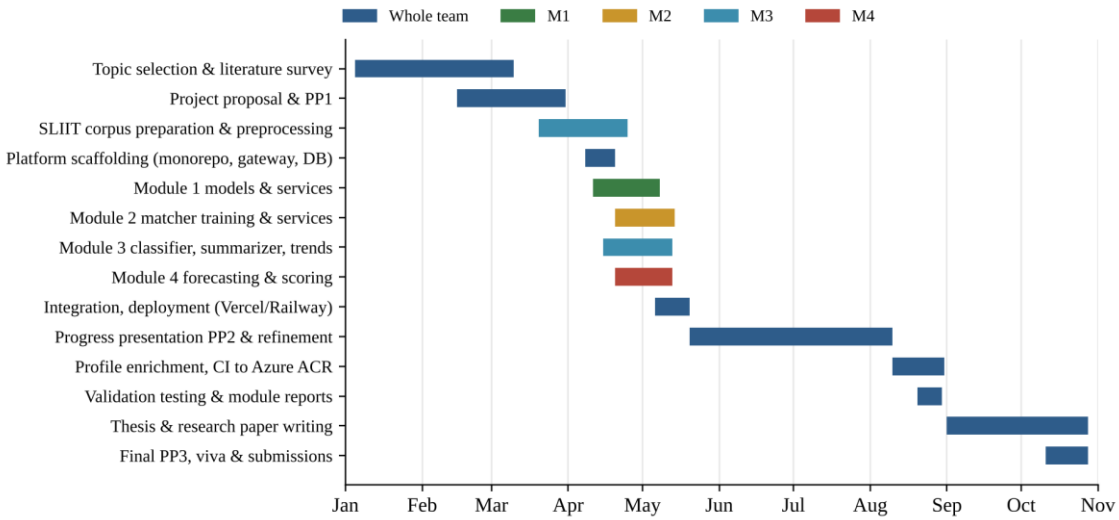


Figure 5.1: Project timeline (January–October 2026)

5.3 Deployment and Operational Configuration

Figure 5.2 shows the deployment pipeline. A push to `main` triggers a GitHub Actions workflow that deploys the front end to Vercel (region sin1, with X-Frame-Options, X-Content-Type-Options and Referrer-Policy headers); Railway's native GitHub integration rebuilds the gateway and the four Dockerised services, with SBERT models and indices baked into images; a further workflow builds images for Azure Container Registry and the paper-chat service can be deployed to Hugging Face Spaces. Configuration (Supabase URL and keys, Gemini, Semantic Scholar and SMTP credentials) is injected as environment variables from `.env.example` templates, never committed. Supabase provides managed backups of the database; monitoring relies on service status endpoints, the dashboard model-status grid and platform logs.

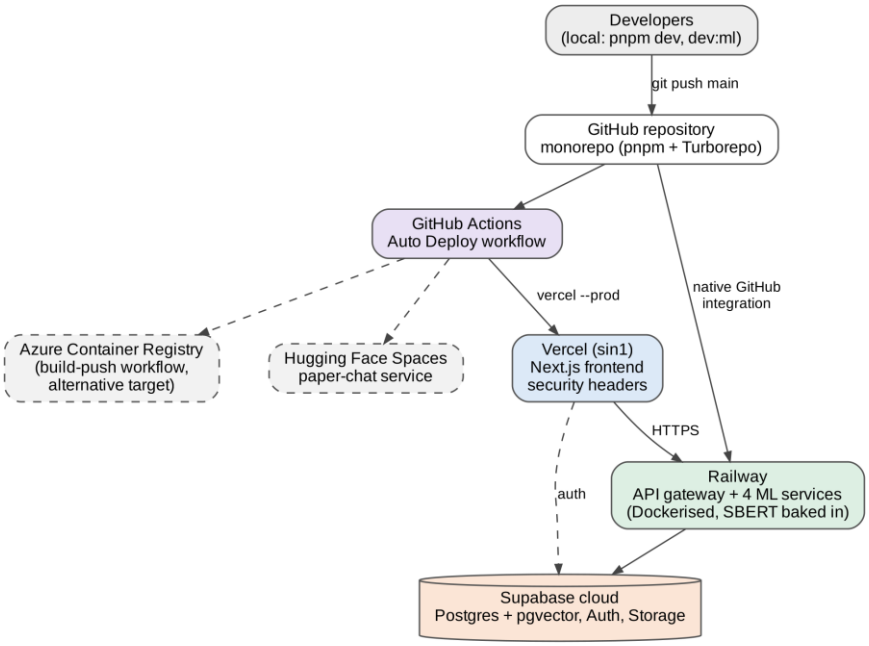


Figure 5.2: Continuous deployment pipeline

5.4 Evidence of Team Execution

The repository contains 80 commits between 10 April and 30 August 2026 with descriptive messages (feature, fix, performance, security, CI), pushed from the integration lead's account; module ownership is recorded in the README team table, in each module report and in the individual theses. Each module has a validation workbook and report (`docs/module-reports`), signed off by the supervisor, and reproducible test scripts. MS Planner boards, MS Teams meeting records and individual research diaries are maintained separately and are presented at the final assessment, as required by the module.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

Evaluation followed the IT4010 distinction between *evaluation* (how well does it perform?) and *validation* (does it solve the intended problem?). Success criteria were fixed in the proposals: component-level accuracy targets (Table 6.1), functional correctness of every feature, integration through the gateway, and supervisor validation. Evaluation used held-out model metrics, offline benchmarks on the corpus, back-tests and live functional tests executed against the running services on 28 August 2026.

6.2 Integration and End to End Testing

- **Unit and component level:** model training logs and held-out metrics; offline benchmarks (3,866 citations; 600 abstracts; ARIMA hold-out).
- **Interface and system level:** 52 live API test cases across the four services (13 + 13 + 14 + 12), each with input, criteria, actual output, verdict, severity and root cause; a smoke-test script that calls every gateway route; Playwright end-to-end specifications for each module.
- **Regression:** defects found in validation were fixed and re-tested (for example, the IEEE citation-shape failure fixed on 29 August 2026).
- **Security and performance:** authentication and rate limits verified at the gateway; latency measured for core paths (citation parsing 0.08 ms, quality scoring 1.6 ms, success prediction 0.4 ms).
- **User acceptance:** supervisor review and sign-off of all four validation reports; task-based student UAT protocols prepared for the final assessment (individual theses, Appendix D).

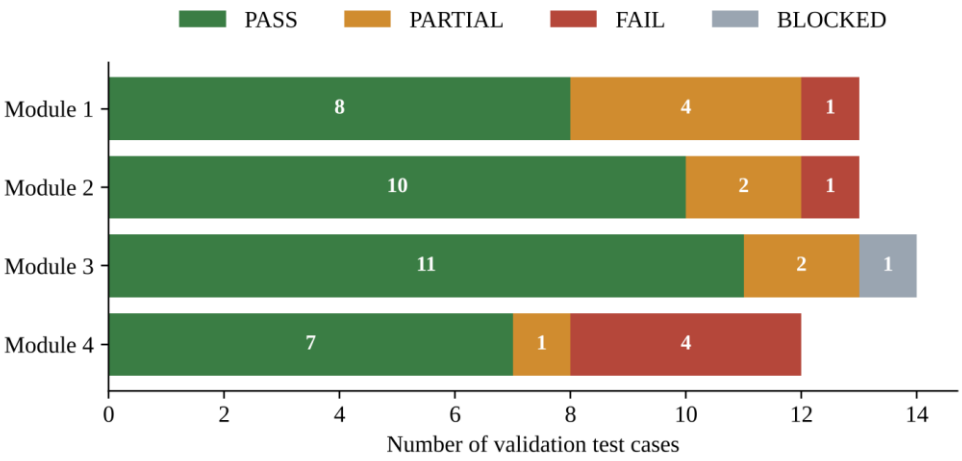


Figure 6.1: Live functional validation outcomes for the four services (52 test cases)

Across the 52 cases (Figure 6.1), 36 passed, 9 were partial, 6 failed and 1 was blocked by an unpopulated table in the test environment. Failures were concentrated in one Module 4 issue (a partial-year growth baseline and its insight text) and in Gemini parse failures; no failure was architectural.

6.3 Overall Results

Table 6.1: Synthesised component results against targets

Component	Key result	Target	Outcome
M1 Citation parser	Field accuracy on 3,866 refs: year 100%, title 98.0%, surnames 90.9%; 0.077 ms	≥ 90%	Met
M1 Proposal generator	All exemplars on topic; every section traceable	Grounded drafts	Met
M1 Gap analysis	Relevant for covered topics; no confidence gate	3–5 relevant gaps	Partly met
M2 Supervisor matcher	Accuracy 0.946, precision 0.941, recall 1.000, F1 0.970 (147 pairs)	Acc ≥ 85%, F1 ≥ 0.75	Met (optimistic)
M2 Aspect sentiment	Correct for clear cases; neutral collapse on mixed text	Differentiated aspects	Partly met
M3 Corpus quality	Completeness 0.704, consistency 0.913, duplicates 10.6%	≥ 95% compliance	Not met
M3 Topic classifier	Macro-F1 0.054; top-3 accuracy 0.692	Macro-F1 ≥ 0.80	Not met
M3 Summariser / comparator	Presets respected; paraphrase 0.987 → high; unrelated → minimal	Faithful; correct extremes	Met
M4 Forecasting	Hold-out 1 − MAPE 84.7% (general); worse than naïve for 4 of 6 topics	1 − MAPE ≥ 78%	Partly met
M4 Quality / success	Discriminates research vs non-research text; R^2 and AUC inflated by leakage	F1 > 0.75, AUC > 0.80	Met on proxy labels

The shared corpus underpins these results; Figure 6.2 shows its distribution by year, including the partial 2026 year that affected Module 4's growth calculation.

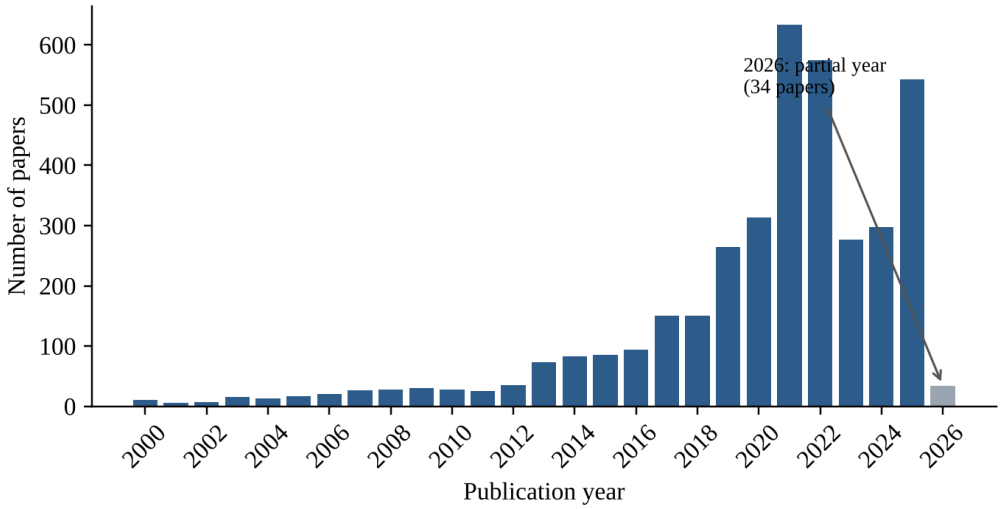


Figure 6.2: Shared SLIIT corpus by publication year

Figures 6.3 to 6.6 show the integrated platform in operation, one screen per module, rendered from responses recorded in the validation run.

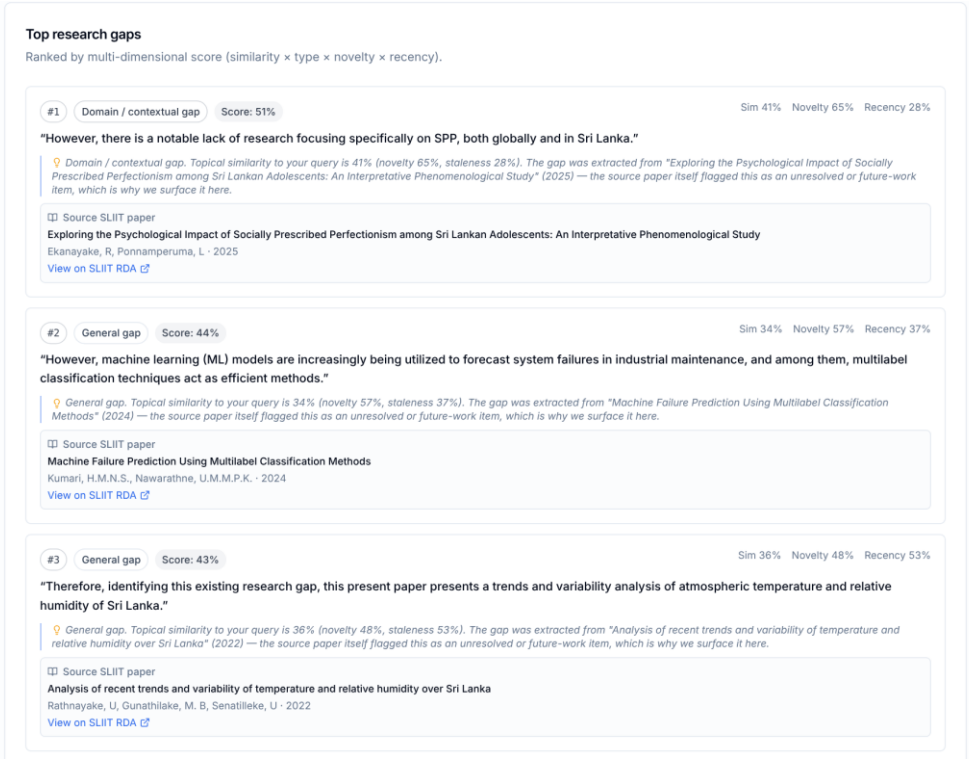


Figure 6.3: Module 1: research gaps grounded in cited SLIIT papers

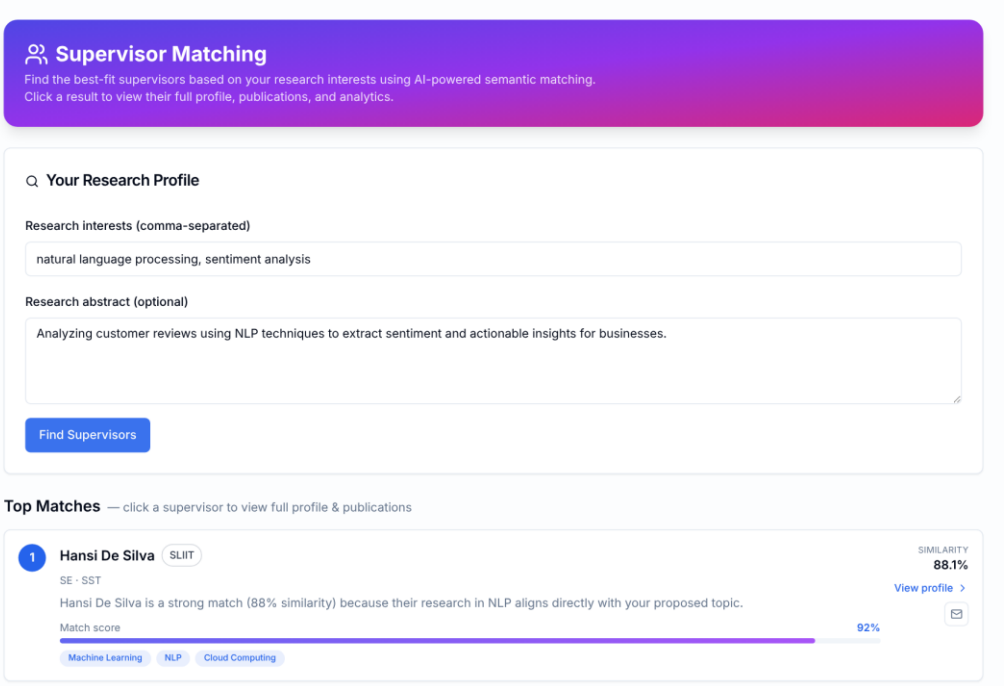


Figure 6.4: Module 2: ranked and explained supervisor matches

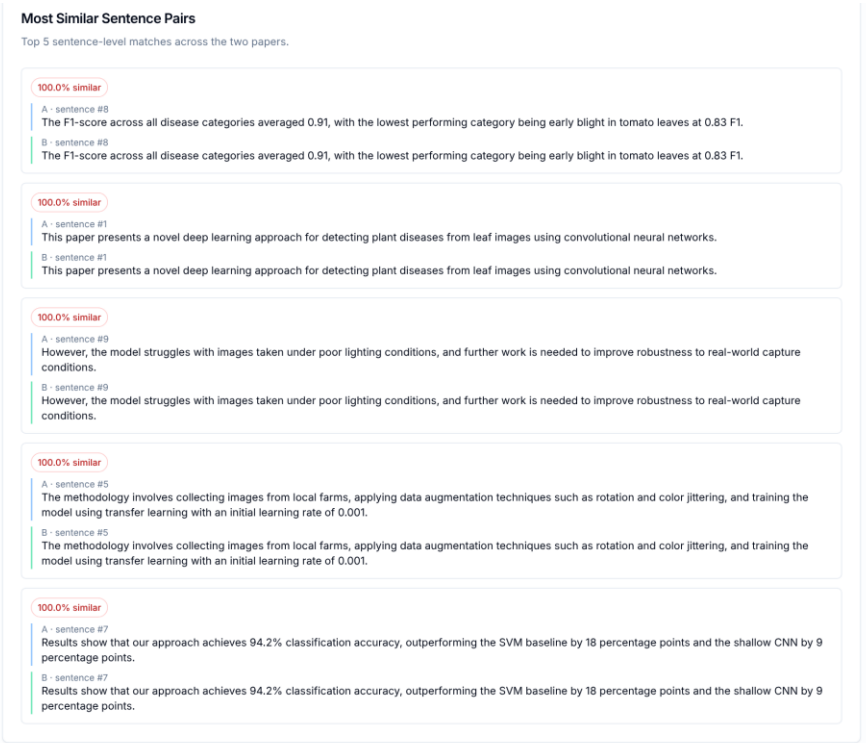


Figure 6.5: Module 3: pairwise plagiarism comparison with matched sentence pairs

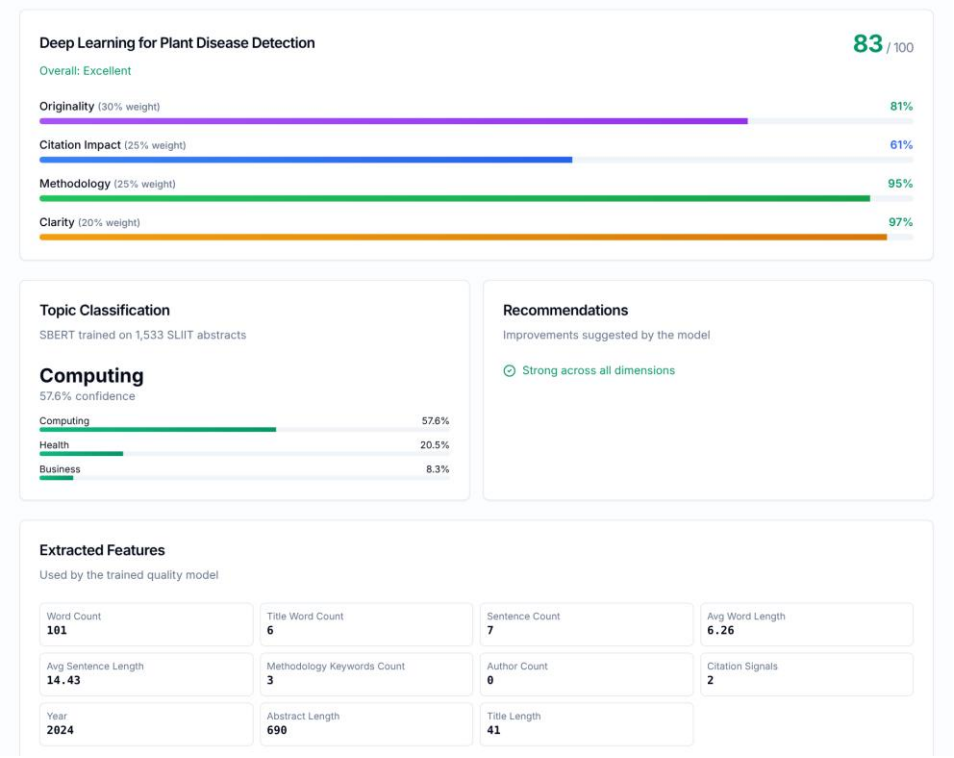


Figure 6.6: Module 4: quality scoring with dimension breakdown and extracted features

6.4 Stakeholder and User Validation

The project supervisor, Ms. Sanjeevi Chandrasiri, reviewed the validation workbook and report for each module, including all 52 cases with their evidence, and signed them off with the comment that the outputs had been reviewed, validated and confirmed. The principal findings she highlighted were the strength of retrieval-based features (similar papers, proposal grounding, supervisor matching, summarisation and comparison) and the priority issues to address (citation-shape defects, gap confidence, sentiment parsing, classifier bias and growth baseline). Changes made in response include the IEEE citation fix and the documentation of all remaining issues. Student user-acceptance testing with the prepared task protocols and the System Usability Scale [38] is scheduled for the final assessment period; its results are not yet available and are not claimed here.

6.5 Achievement of Common Objectives

Table 6.2: Achievement of group objectives

Objective	Status	Justification
GO1 Single authenticated platform	Fully achieved	All features reachable through one gateway and UI
GO2 Shared embedded corpus	Fully achieved	4,219-record index reused by Modules 1, 3 and 4
GO3 Grounded recommendations	Fully achieved	Gaps, proposals, related papers and matches cite real SLIIT sources
GO4 Component accuracy targets	Partly achieved	Met for citations, matching, comparison; not for classifier or corpus quality
GO5 Live validation with sign-off	Fully achieved	52 cases; four reports signed off
GO6 Continuous CPU-only deployment	Fully achieved	Vercel and Railway via GitHub; images with baked models

6.6 Integrated Limitations and Threats

- **Data dependencies:** corpus duplicates (10.6%) and the partial 2026 year propagate to Modules 1, 3 and 4.
- **Generative fragility:** Gemini JSON-parse failures degrade silently in two places.
- **Model deployment:** fine-tuned and graph models are not shipped through a model registry, so production may run base models or fallbacks.
- **Evaluation validity:** several metrics are optimistic (template-based citation benchmark, overlapping matcher evaluation, leaked quality labels); student UAT is pending.
- **Generalisability:** the corpus and supervisor directory are SLIIT-specific and computing-dominated.

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

For **students**, Researchly AI replaces four or five separate tools with one platform whose outputs are grounded in their own institution's research; the proposals estimated time savings of 15–20 hours per project for students and 10–15 hours per research cycle for supervisors, figures that remain to be measured. For **supervisors and coordinators**, it offers better-matched students, grounded proposals, early warning and a transparent view of supervision quality. For **institutions**, it turns a repository into research infrastructure and supports academic integrity.

7.2 Market Alternatives and Differentiation

Table 7.1: Market alternatives and differentiation

Alternative	Strength	Weakness relative to Researchly AI
Reference managers (Zotero, Mendeley)	Mature citation management	Manual import; no gaps, proposals or matching [39]
Similarity checkers (Turnitin)	Large comparison database	Post-submission; per-document; no guidance [40]
Research analytics suites (Scopus/SciVal, VOSviewer)	Global bibliometrics	Descriptive, citation-centric, subscription-based [10]
General LLM assistants	Fluent, broad	Ungrounded; unverifiable citations and gaps [13]
Staff directories / spreadsheets	Familiar	No semantic matching or analytics

The defensible differentiation is the combination of an institution-specific evidence base, integration across the research workflow and explainability, which together create a data network effect: every deposited paper improves every module.

7.3 Business Model and Feasibility

The proposed model is staged: a free pilot at SLIIT in year one; a freemium model in year two (core features free; premium analytics, gap analysis and comparison reports at roughly LKR 400–600 per user per month); and institutional licences in year three

(LKR 150,000–200,000 per institution per year) with corpus onboarding and support. Channels are university IT and library partnerships. Technical feasibility is demonstrated by the deployed system; operational feasibility by CPU-only inference, which keeps hosting costs within the LKR 120,000–150,000 per year estimated in the proposals, with Gemini used only as a fallback or for cached summaries.

7.4 Adoption, Intellectual Property and Compliance

- **Adoption barriers:** trust in AI outputs (addressed by grounding and explanations), integration with existing repositories (DSpace API supported), and staff concerns about effectiveness scores (addressed by transparency and advisory use).
- **Intellectual property:** platform code, ranking and blending formulas, datasets compiled by the team and integration design belong to the project team and SLIIT under the module's rules; third-party libraries are used under open-source licences; corpus metadata remains attributed to its source.
- **Data protection and ethics:** PDPA-aligned processing [30], RLS-based access control [21], OTP-verified ratings, private uploads, advisory rather than decisive use of scores, and disclosure of AI assistance.
- **Professional responsibility:** known limitations are documented, and features with unvalidated outputs are not presented as production-ready.

7.5 Sustainable Development Goal Contribution

The plausible contribution to SDG targets 4.3 and 4.4 is wider and fairer access to research support; to target 9.5, stronger institutional research infrastructure. Possible negative effects include over-reliance on automated suggestions, homogenisation of topics toward what the corpus already contains, and reputational harm from misinterpreted scores. Measures to sustain benefits include keeping humans in the loop, displaying sources and uncertainty, periodic bias reviews of matching and scoring, and longitudinal measurement of student outcomes [12].

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

The common problem was that students and institutions lack an integrated, explainable research-support system grounded in the institution's own output. Researchly AI addresses it with a deployed platform in which four modules share one gateway, one schema and one embedded corpus. The integrated evidence shows that the grounding principle works: citations, gaps, proposals, related papers and supervisor matches are traceable to real SLIIT sources and were validated by the supervisor. It also shows where integration magnifies weaknesses: data quality in the shared corpus, empty upstream tables and silent generative fallbacks affect several modules at once.

8.2 Principal Integrated Contribution

Beyond the four components, the team produced (i) a shared, audited and embedded institutional research corpus that any future SLIIT service can reuse; (ii) an integration architecture — gateway, shared library, fallback tiers, status endpoints and continuous deployment — that lets independently developed AI services operate as one product on CPU-only infrastructure; and (iii) a validated, honest picture of what institution-grounded AI research support can and cannot yet do.

8.3 Future Directions

1. Populate and de-duplicate the shared corpus and enrich DOIs.
2. Introduce a model registry so that fine-tuned and graph models ship with every deployment and responses report active model versions.
3. Make generative failures visible and retryable; add confidence gates (gap analysis) and partial-period handling (forecasting).
4. Collect expert labels (proposal quality, feedback aspects, supervisor fit) to replace proxy labels and measure agreement.
5. Complete student UAT with SUS and measure time savings and outcome changes over a full project cycle.
6. Pilot with other Sri Lankan universities through the DSpace connector.

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APPENDICES

Appendix A: Integrated Interface Specification

All routes are exposed by the gateway under `/api/v1/` and require a Supabase JWT.

Table A.1: Module 1 endpoints (services/module1-integrity, :8001)

Method and path	Purpose
POST /citations/parse	Parse raw citation
POST /citations/format	Format record (APA/IEEE)
POST /citations/lookup-doi	CrossRef DOI lookup
POST /citations/lookup-title	CrossRef title search
POST /citations/in-text	In-text citation
POST /citations/reference-list	Reference list
POST /citations/similar-papers	Similar SLIIT papers
POST /gaps/analyze	Gap analysis (topic)
POST /gaps/analyze-pdf	Gap analysis (PDF)
POST /gaps/analyze-full-paper	Gap analysis (text)
POST /gaps/report	HTML report
GET /gaps/status	Index status
POST /proposals/generate	Proposal generation
GET /proposals/status	Index status
POST /mindmaps/generate	Mind map

Table A.2: Module 2 endpoints (services/module2-collaboration, :8002)

Method and path	Purpose
POST /matching/supervisors	Supervisor matching
GET /matching/supervisors/{id}/papers	Profile, publications, focus
POST /matching/groups	Create group
GET /matching/groups	List groups
GET /matching/groups/{id}	Group detail
POST /matching/groups/{id}/join-request	Join request
POST /matching/peers	Legacy peer matching
POST /feedback/request-otp	Request OTP
POST /feedback/verify-otp	Verify OTP
POST /feedback/analyze	Aspect sentiment
GET /feedback/supervisors	Rateable directory
POST /feedback/submit	Submit rating
GET /feedback/by-supervisor	Ratings for supervisor

Method and path	Purpose
GET /effectiveness	Effectiveness list
GET /effectiveness/by-key	Effectiveness detail
GET /effectiveness/{id}	Legacy detail

Table A.3: Module 3 endpoints (services/module3-data, :8003)

Method and path	Purpose
POST /data/categorize	Categorisation
GET /data/categorize/status	Model status
POST /data/summarize	Summarise text
POST /data/summarize/upload	Summarise PDF
POST /data/summarize/report	HTML report
GET /data/summarize/status	Model status
GET /data/plagiarism-trends	Legacy trends
POST /data/plagiarism-trends/search	Trend search
POST /data/plagiarism-trends/search/report	HTML report
POST /data/plagiarism-trends/compare	Compare texts
POST /data/plagiarism-trends/compare-pdf	Compare PDFs
POST /data/plagiarism-trends/compare/report	HTML report
GET /data/plagiarism-trends/status	Model status
GET /data/quality	Quality metrics

Table A.4: Module 4 endpoints (services/module4-analytics, :8004)

Method and path	Purpose
GET /analytics/trends	Per-topic forecasts
POST /analytics/trends/compare	Compare topics
GET /analytics/trends/insights	Emerging topics and recommendations
GET /analytics/trends/topics	Available topics
POST /analytics/trends/report	HTML report
POST /analytics/quality-score	Quality scoring
POST /analytics/papers/analyze-text	Quality analysis (text)
POST /analytics/papers/upload	Quality analysis (PDF/TXT)
POST /analytics/predict	Success prediction
POST /analytics/predict/upload	Success prediction (file)

Appendix B: Team Contribution Declaration

The members confirm that the contributions summarised in Table 4.3 and detailed in the individual theses are an accurate record of individual and shared work.

Table B.1: Team contribution declaration

Member	Student ID	Component	Signature	Date
Kariyawasam K.D.T.	IT22210524	Module 1		
Gunathilaka S.P.U.	IT22055194	Module 2		
Hewamanne N.V.	IT22135902	Module 3		
Jayasundara H.W.S.S.	IT22553096	Module 4		

Appendix C: Academic Integrity and AI-Use Disclosure

Table C.1: AI-use disclosure

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic) — AI coding assistant	Debugging and development suggestions; execution of module validation harnesses and drafting of module reports (28 Aug 2026)	Every test re-run and evidence inspected; supervisor sign-off	Chapters 5, 6
Claude (Anthropic)	Drafting and formatting this report from repository evidence; diagrams and charts generated from repository data	Numbers cross-checked with repository files; references checked	All chapters
Google Gemini 2.5 Flash	Runtime component (mind maps, fallbacks, sentiment, summaries)	Outputs treated as untrusted; fallbacks implemented	Chapter 3

Appendix D: Platform Screenshots

Screenshots were captured from the production build of the Researchly AI front end; module screens render responses recorded from the live services during the 28 August 2026 validation run.

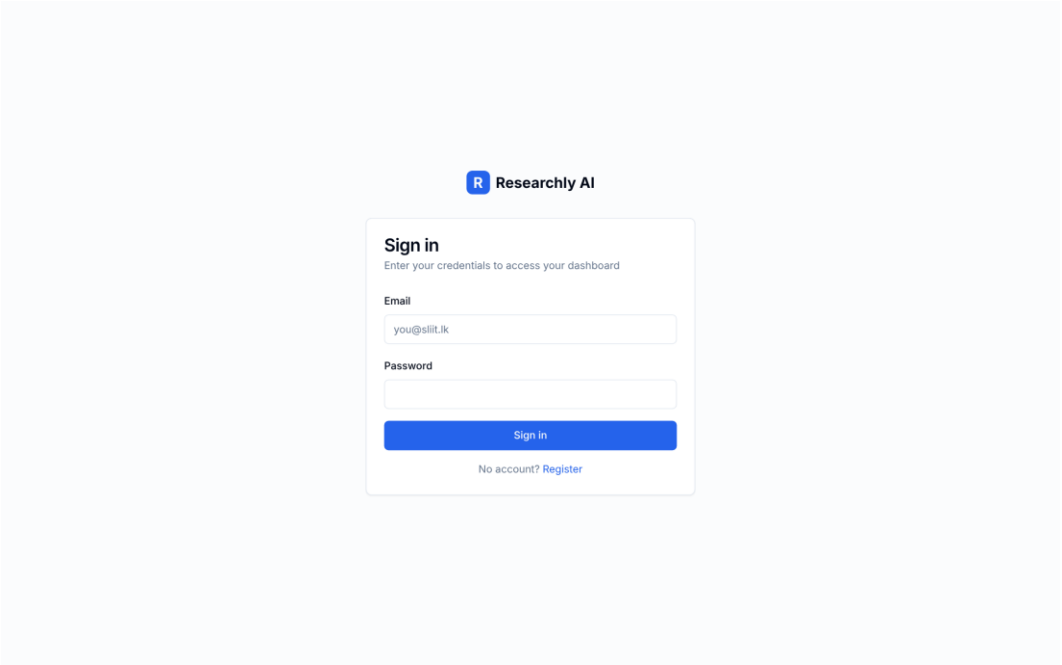


Figure D.1: Sign-in page (Supabase authentication)

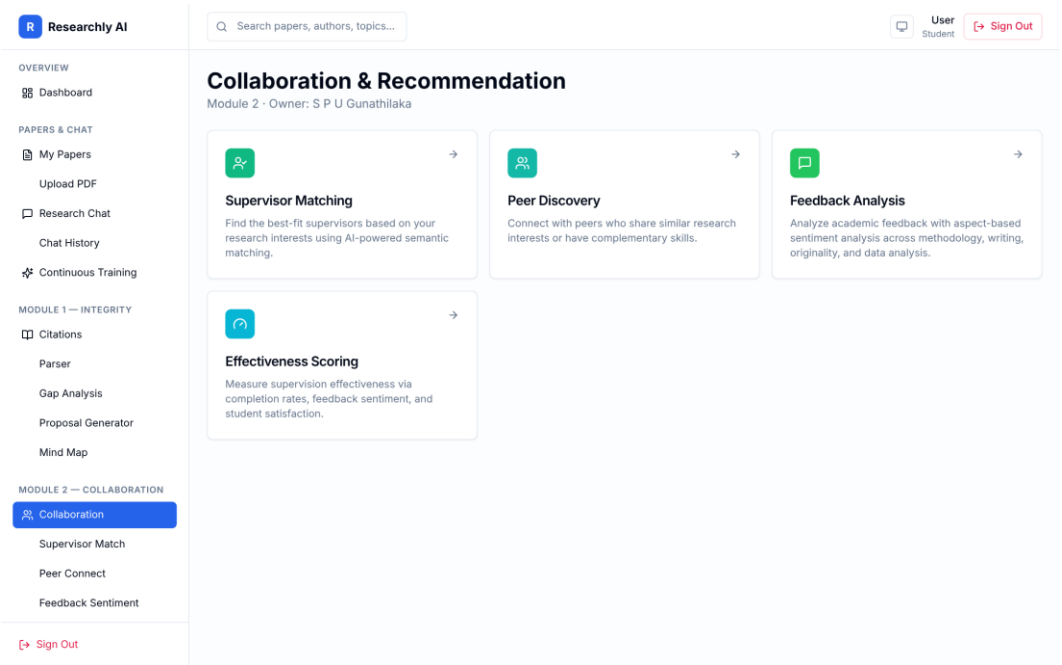


Figure D.2: Module 2: Collaboration and Recommendation overview page

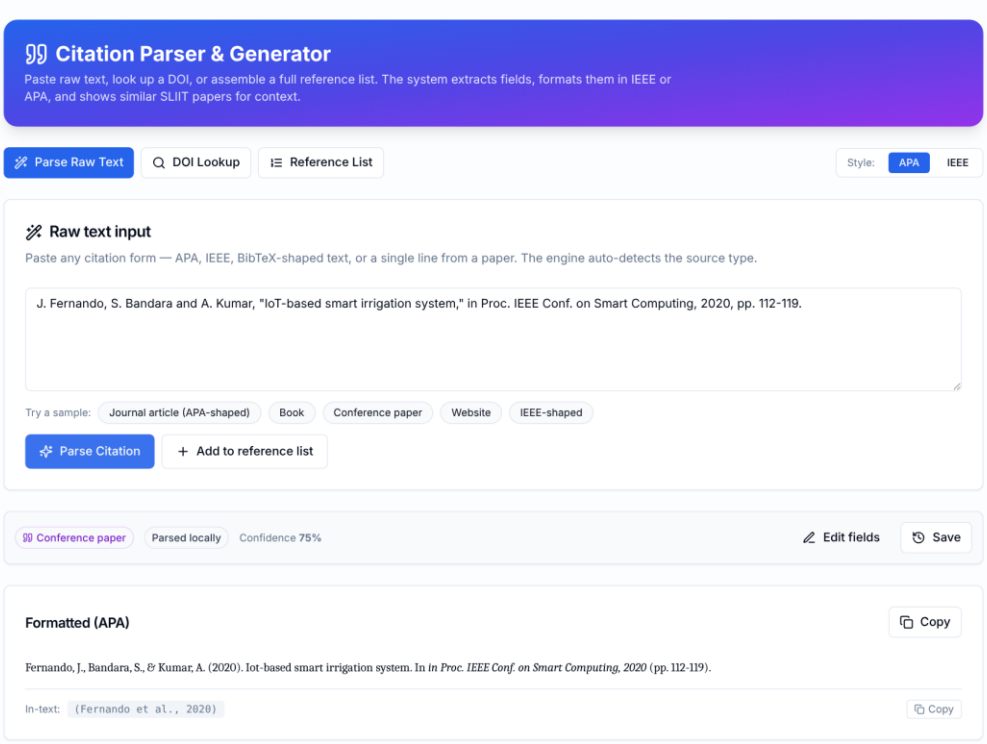


Figure D.3: Module 1: citation parser with APA output and confidence

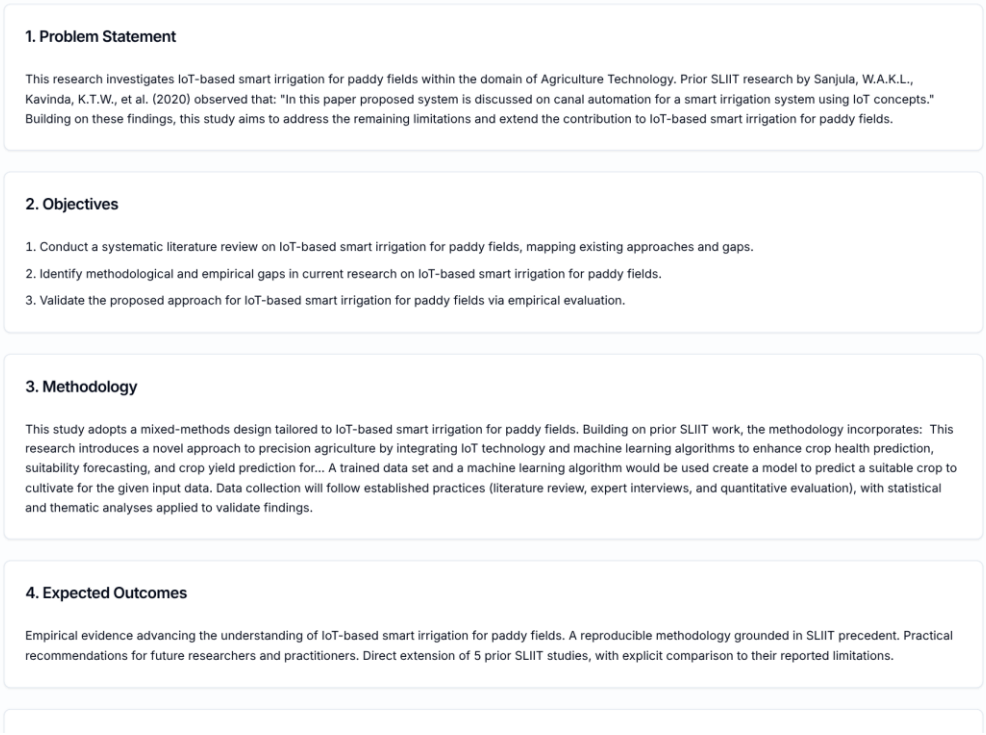


Figure D.4: Module 1: grounded proposal draft

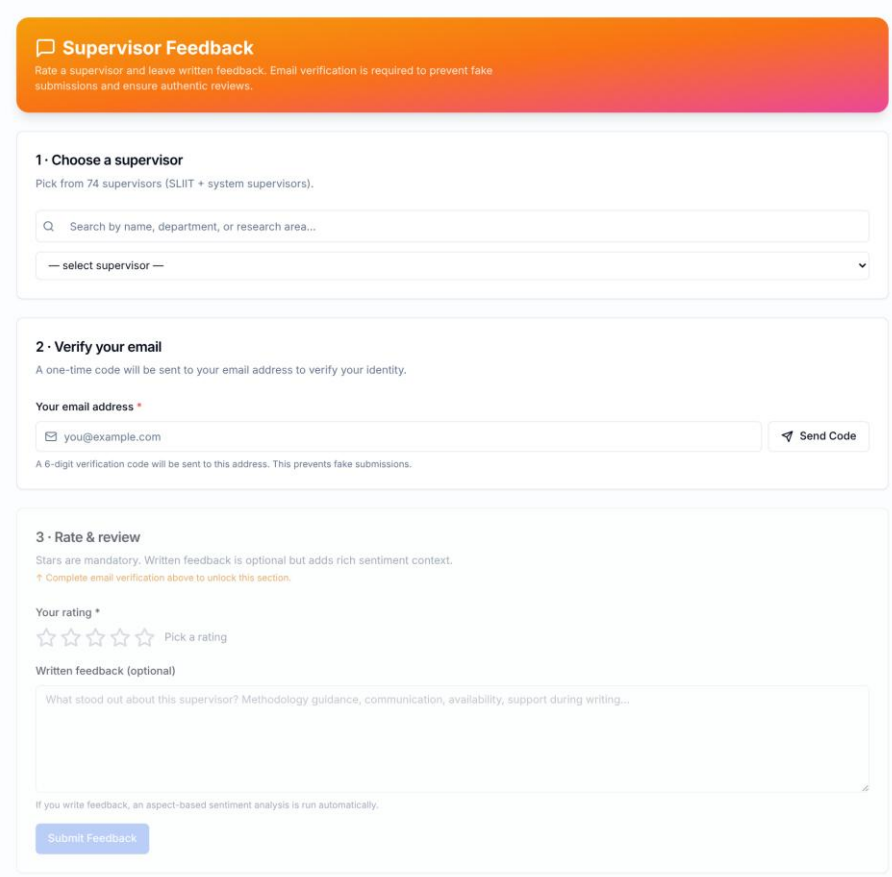


Figure D.5: Module 2: verified supervisor feedback workflow

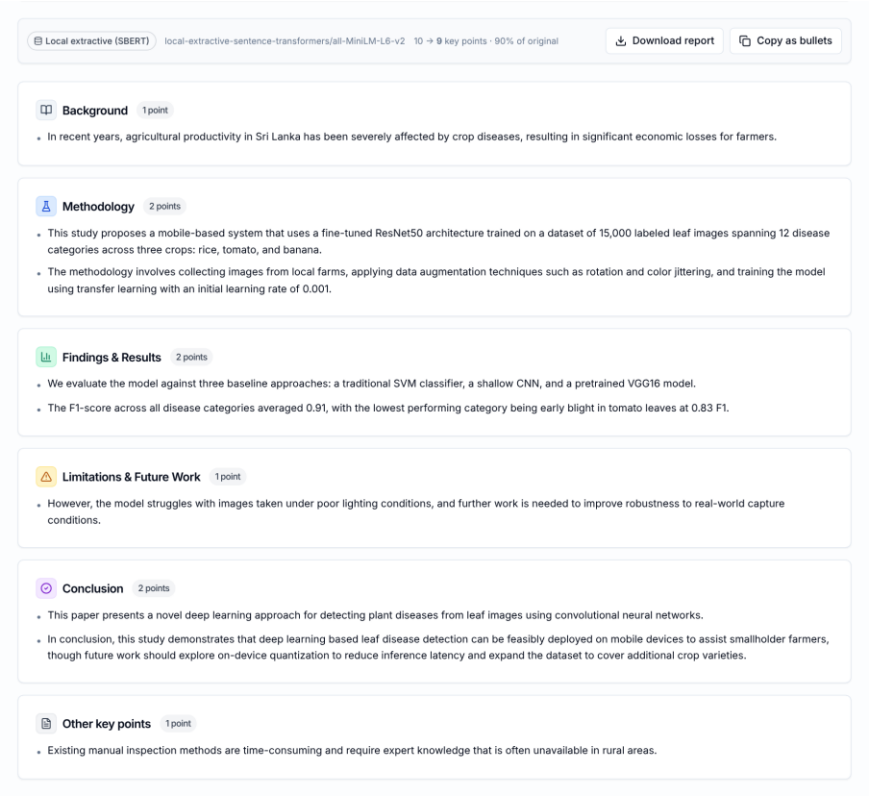


Figure D.6: Module 3: section-grouped extractive summary

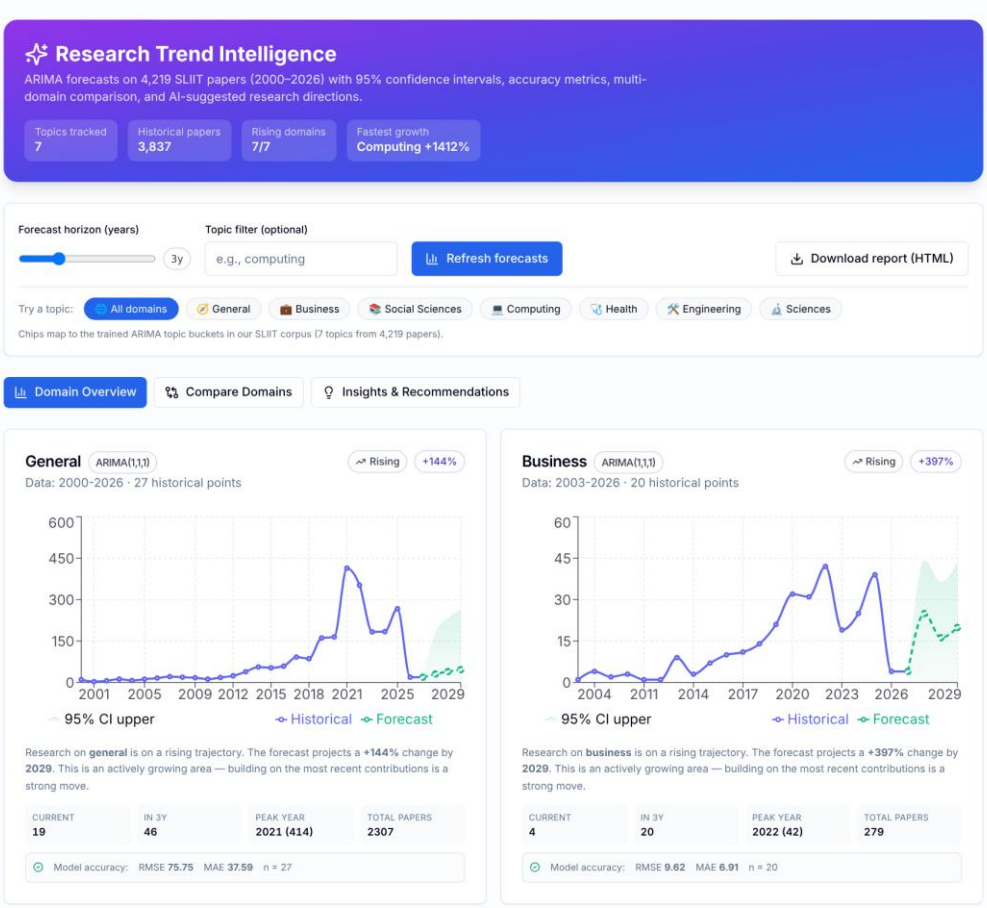


Figure D.7: Module 4: trend forecasts with 95% intervals

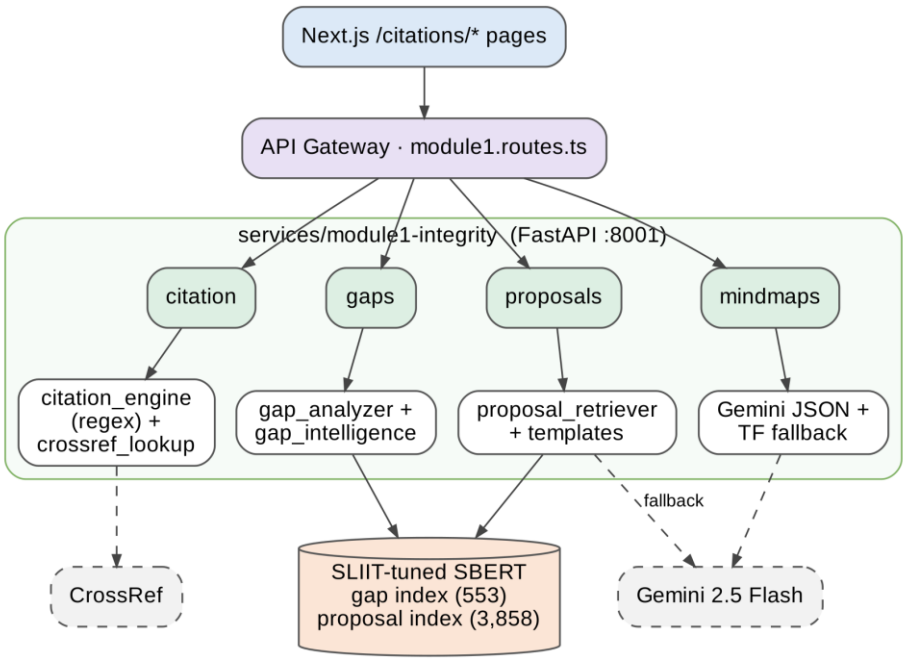


Figure D.8: Module 1 component architecture

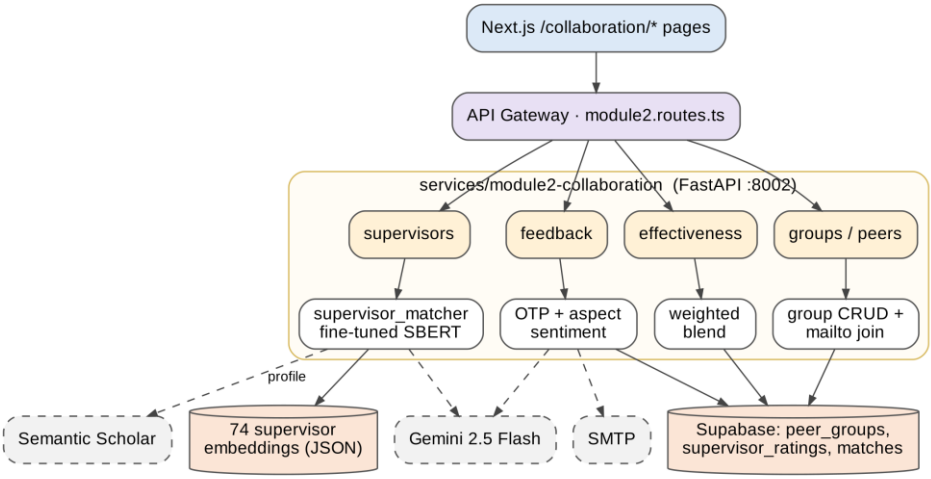


Figure D.9: Module 2 component architecture

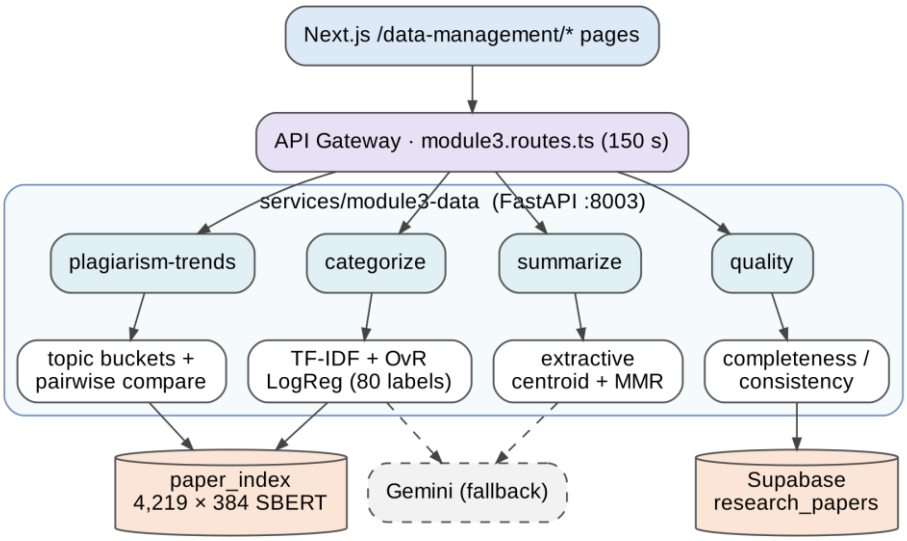


Figure D.10: Module 3 component architecture

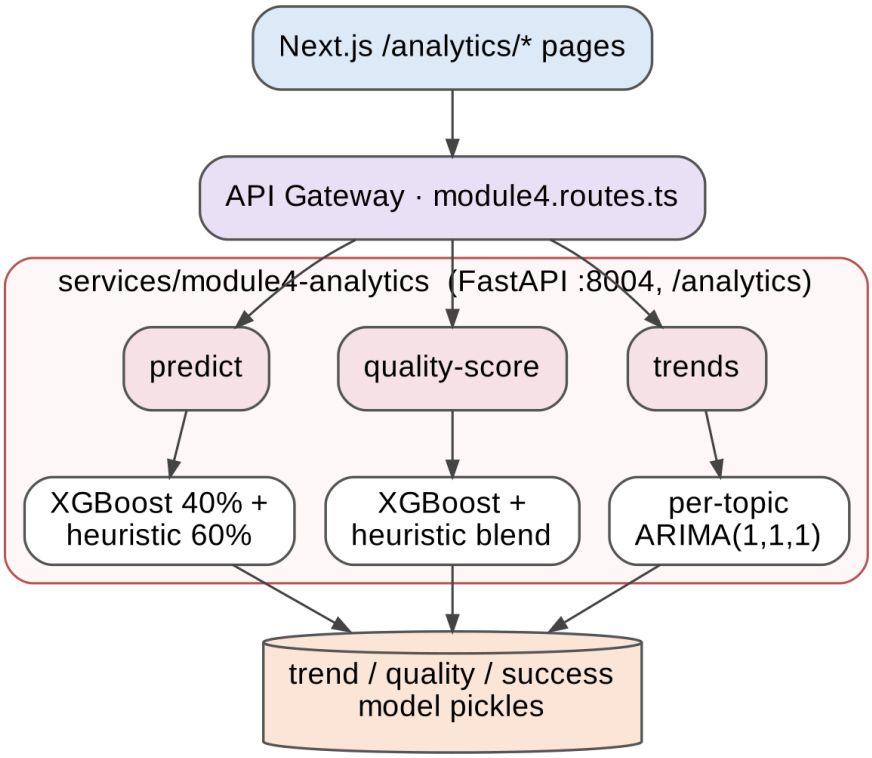


Figure D.11: Module 4 component architecture